# Riti Grover

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## Education

**Manipal Institute of Technology** — Bengaluru, Karnataka
*B. Tech. in Computer Science (Information Technology), CGPA: 8.70* — *Jul. 2024 – May 2028*

**RS Memorial Higher Secondary School** — Katni, Madhya Pradesh
*PCM – Class XII: 81%* — *2023 – 2024*

## Experience

**Research Intern – AI-Driven Edge Computing for IoT** — May 2025
*IIT Roorkee* — *Roorkee, Uttarakhand*

- Researched deep learning architectures, including Transformers, and software engineering methodologies to design scalable, real-time resource-allocation solutions for IoT edge systems
- Designed, built, and tested a hybrid machine learning and optimization framework, benchmarking alternative approaches against standard baselines to recommend the most effective solution

**Creative Intern** — May 2025
*Luscious Leopard* — *Indore, Madhya Pradesh*

- Redesigned website frontend layouts and produced Instagram and website content, improving user experience and aligning visuals with brand identity

## Projects

**Smurfing Hunter** | *Blockchain AML Intelligence, PyTorch Geometric, Flask, D3.js, GNN* | GitHub

- Analyzed transaction data requirements and researched graph-based ML methodologies to design an automated solution for detecting suspicious patterns and generating risk reports
- Designed, coded, and tested a 3-layer Graph Convolutional Network (GCN) to classify wallet behavior, validating detection of three money-laundering typologies: smurfing, peeling chains, and mule networks
- Built a robust D3.js forensic dashboard with interactive Sankey diagrams, reducing manual transaction-analysis time for compliance teams by ~40%
- Developed an automated Suspicious Activity Report (SAR) generator that reliably surfaces the top 50+ high-risk anomalies per batch with real-time risk scoring

**Timetable AI** | *Student Platform, Constraint-Based Scheduling System, FastAPI, Python, JavaScript* | GitHub

- Analyzed scheduling requirements and designed a constraint-driven architecture using Constraint Satisfaction Problem (CSP) methodologies to generate robust, conflict-free timetables under complex dependency constraints
- Coded and tested an optimized CSP solver combining backtracking search, Minimum Remaining Values (MRV) heuristics, and forward checking to improve reliability and reduce computational overhead
- Architected a modular, scalable FastAPI/Python backend with separated solver, graph, and metrics components, paired with a responsive HTML/CSS/JavaScript frontend

## Courses and Certifications

**Data Analytics Foundations** *(DeepLearning.AI)* — certificate
**Fundamentals of Java Programming** *(Board Infinity)* — certificate
**Integrating with Structured Databases** *(IBM SkillsBuild)* — certificate
**Linux on IBM LinuxONE** *(IBM SkillsBuild)* — certificate

## Clubs, Committees and Co-Curricular Activities

**Vice President**, LitSoc – Literature Society | **Social Media Manager**, OtakuSpot – Pop Culture Club | **Executive Member**, IEEE Computer Society | **Organising Committee**, TechSolstice – Technical Fest | **Content Writer**, PCB – Photography Club | **Marketing Team**, Radar – Robotics Club
*MIT Bengaluru – 2024 to 2026*

## Skills

**Technical Skills**: Java, C, SQL, Python, HTML/CSS, Data Analysis, Excel, Data Visualization
**Professional Skills**: Leadership, Team Management, Marketing, Content Strategy, Event Planning, Communication, Public Speaking, Strategic Thinking, Adaptability, Creativity
**Languages**: English, Hindi